Zhiheng (Kevin) Li

li004628@umn.edu | +1 (612) 283-4843 | Minneapolis, MN | github.com/ZhihengLi0

EDUCATION

University of Minnesota, Twin Cities Minneapolis, MN

*Bachelor of Science in **Mathematics** • **Computer Science Minor***

Expected May 2028

- **GPA:** 4.0/4.0 | **Honors:** Dean's List, Fall 2025, Spring 2026 · Tau Sigma Honor Society
- **Relevant Coursework:** Calculus I & II (A), Linear Algebra & Differential Equations (A), Physics for Scientists and Engineers I & II (A), Introduction to Programming Concepts (Python, A), Introduction to Statistical Analysis (A)

RESEARCH EXPERIENCE

SuperCDMS Dark Matter Experiment - School of Physics & Astronomy, University of Minnesota Minneapolis, MN

Undergraduate Research Assistant & UROP Fellow, Advisor: Prof. Yan Liu Oct 2025 - Present

• Built **cdms_spotlight**, a multilingual natural-language query system for the SuperCDMS DQM database hosted at **SLAC**: server-side filtering scaling from ~1,300 to ~100,000 series, cuts on any of 10,929 ODB parameters per series, self-learning local NLP (TF-IDF + logistic regression); runs 24/7 as a Slack bot in the official collaboration workspace, pip-installable CLI; presented at the SuperCDMS Software Meeting on July 14, 2026; contributed to the follow-up DQM/ODB upgrade, prompted by cdms_spotlight, that moves non-metadata parameter cuts server-side so queries no longer pull data to the client. Jul 2026

SNOLAB Run 4 Phonon Pulse Analysis May–Sep 2026

- Built the **phonon pulse template pipeline** for 13 SNOLAB R4 detectors (12 phonon channels each): event selection around the 10.37 keV activation line (~120 GB raw traces), free-pretrigger 2-exponential fits with quality cuts; the 1×1 and PCA N×M ROOT templates serve as the R4 pulse templates for optimal-filter reprocessing; presented at the SuperCDMS Analysis Working Group meeting on July 29, 2026. May–Jul 2026
- Measured the **phonon collection efficiency** of detector Z7: converted every K-line pulse from current to power with the TES electrothermal equations and integrated the fitted pulse in closed form, validated against the official absorbed-energy RQ; presented at the SuperCDMS Analysis Working Group meeting on September 23, 2026. Jul–Sep 2026

• Built **BlueFors-Alert**: minute-level PostgreSQL sync of the 64 GB CS2 database to a Raspberry Pi; Slack monitor with dual IDLE/COLD thresholds and time-range plots of any sensor; extended to a liquid-nitrogen scale. Jul 2026

• Extended RuOx (R31279 & R31839) **calibration curves** to 5 mK–300 K (PCHIP + Mott VRH); the generated .340 files are now in use on the lab's BlueFors dilution refrigerator. May 2026

• Built a **Python pipeline** for LED calibration data from 11 cryogenic Ge channels (5,200+ events, 10+ GB raw): baseline subtraction, SNR-based glitch detection, peak alignment, exponential fitting. Oct 2025

Mouse Eye Tracking for Neural Alignment - Deisseroth & Mitra Labs, Stanford University Remote

Research Collaborator (remote), with Kaiwen Sheng, Ph.D. student in Bioengineering Aug 2026 - Present

• Built a reproducible **DeepLabCut (PyTorch) pipeline** that tracks eight keypoints on the mouse eye (pupil ellipse endpoints, eyelids, corners) in ~57,000-frame videos and derives pupil size, eye opening and blinks for alignment with brain recordings; code, audited aggregate results and a bilingual report page are public on GitHub. Aug–Sep 2026

• Measured **how many human labels a new video needs**: pre-label with the previous video's model, correct, retrain, stop when two 20-label steps improve the error by ≤ 3%; over 19 videos from 3 mice, 16 videos plateaued at ≤ 40 labels (median per-frame error 3–25 px on frozen test sets, 820 labels in total). Sep–Oct 2026

• Currently developing a **label-free time-series detector for unreliable frames** and a repair rule that changes only the flagged frames, so that human correction stays minimal and no temporal smoothing enters the signal. Oct 2026

RESEARCH & INDUSTRY EXPERIENCE

Low-Carbon Transition & Industrial Monitoring Platform **Beijing / Shanghai, China**

Faculty-led Researcher (research phase) → Algorithm Engineer (industry phase), Shanghai Huaruizhongxin Technology Co., Ltd.

Oct 2024 - May 2025; Aug 2026 - Sep 2026

• Contributed to a faculty-led low-carbon transition project for China's steel industry, supporting a proposal and implementation plan for national ministry consideration through data collection, database construction, and quantitative analysis.

• During the industry implementation phase, contributed to an energy and carbon-emissions monitoring platform by designing a Python prototype for industrial time-series data quality governance, translating meter/point metadata and quality flags into rules for cumulative, instantaneous, consumption, enumeration, and statistical measurements.

• Implemented historical and real-time cleaning for range, constant, trend, missing-data, and communication anomalies; used same-time historical profiles for outage imputation and cumulative-to-consumption attribution; validated multi-day data with plots and regression tests, then packaged the work with Git version control.

SELECTED PROJECTS

College Students Entrepreneurial Project & Entrepreneurship Roadshow Beijing, China

Team Leader Oct 2024 - Jun 2025

- Led a student team to design **a billiards rest** that combines the high and low bridge heads into one;
- Took it from CAD model to 3D-printed prototypes, iterating through player testing and market research; received **national project approval with seed funding**; **Chinese utility patent** under CNIPA review.
- Invited to present at the **Tsinghua X-Lab & Kering Entrepreneurship Roadshow** to industry leaders.

AWARDS & HONORS

- **UROP** (Undergraduate Research Opportunities Program) Fellowship, University of Minnesota, Summer 2026
- **1st Prize**, Beijing University Student Innovation & Entrepreneurship Competition (Science & Technology Track)
- **1st Prize**, National College E-Commerce "Innovation, Creativity & Entrepreneurship" Challenge, Campus Level | Team Leader
- **3rd Prize**, China International College Students' Innovation Competition, Campus Level (2025) | Team Leader
- **2nd Prize**, NECCS National English Competition for College Students

ACTIVITIES & SERVICE

- **Special Olympics USA Games Minnesota 2026**, Tennis Division - Volunteer Ball Retriever | June 25, 2026
- **Chinese Student Union**, UMN - Active Member | Oct 2025 - Present
- **Chinese Students & Scholars Association**, UMN - Volunteer Calculus I Tutor | 2025 - Present
- **College Table Tennis Club / Football Club / Guitar Club** - Team Member | 2023 - 2024

PROFESSIONAL SKILLS

- **Techniques:** Cryogenic instrumentation, signal processing, data pipeline development, thermometer calibration curve fitting, real-time PostgreSQL sync, local NLP, Slack bot development, waveform template construction
- **Tools:** Python (NumPy, Pandas, SciPy, Matplotlib), scikit-learn, PostgreSQL, Java, ROOT, Git, Linux, Slurm, MSI HPC (Agate)
- **Languages:** Mandarin Chinese (Native), English (Fluent; IELTS 7.0)